

MARKETING CAMPAIGN PERFORMANCE ANALYSIS USING POWER BI

1. Introduction

Marketing campaigns play a crucial role in helping businesses increase brand awareness, attract potential customers, generate leads, and improve revenue. However, running multiple campaigns across different channels requires continuous monitoring to ensure marketing budgets are being utilized effectively.

Marketing analytics helps organizations evaluate campaign performance by measuring key metrics such as impressions, clicks, leads, conversions, revenue, and return on investment (ROI). By analyzing these metrics, businesses can identify successful campaigns, understand customer engagement, and make informed decisions for future marketing strategies.

This project analyzes a marketing campaign dataset inspired by Nykaa's marketing activities. Using Microsoft Excel and Microsoft Power BI, the dataset was cleaned, analyzed, and transformed into an interactive dashboard that provides valuable insights into campaign performance and customer engagement.

2. Problem Statement

Organizations invest significant amounts in digital marketing campaigns across various platforms. However, without proper analysis, it becomes difficult to determine which campaigns generate the highest returns and which marketing strategies require improvement.

The objective of this project is to develop an interactive dashboard that enables marketing managers to:

- Monitor overall campaign performance.
- Evaluate campaign effectiveness.
- Compare campaign types.
- Measure customer engagement.
- Identify high-performing customer segments.
- Support data-driven marketing decisions.

3. Objectives

The main objectives of this project are:

- To analyze marketing campaign performance using business intelligence techniques.
- To clean and prepare the dataset for visualization.
- To calculate key marketing KPIs using DAX measures.
- To build an interactive dashboard using Microsoft Power BI.
- To identify high-performing campaigns and customer segments.
- To provide business recommendations for improving marketing effectiveness.

4. Dataset Description

The dataset contains information about multiple marketing campaigns conducted across different channels and customer segments.

Each record represents a marketing campaign and includes campaign characteristics, customer targeting information, campaign performance metrics, and financial outcomes.

Dataset Overview

- Number of Records: **55,556**
- Number of Columns: **16**

Important Variables

Variable	Description
Campaign_ID	Unique identifier for each campaign
Campaign_Type	Type of marketing campaign (Email, SEO, Paid Ads, Social Media, Influencer)
Target_Audience	Target audience for the campaign
Duration	Campaign duration in days
Channel_Used	Marketing channels used
Impressions	Number of users reached
Clicks	Number of clicks received
Leads	Leads generated
Conversions	Successful conversions
Revenue	Revenue generated
Acquisition_Cost	Cost incurred for the campaign
ROI	Return on Investment
Language	Campaign language
Engagement_Score	Customer engagement score
Customer_Segment	Customer category
Date	Campaign execution date

5. Tools and Technologies Used

Microsoft Excel

Excel was used for:

- Data cleaning

- Data validation
- Duplicate checking
- Missing value analysis
- Preparing the dataset before importing into Power BI

Microsoft Power BI

Power BI was used for:

- Data modeling
- DAX calculations
- KPI development
- Interactive dashboard creation
- Business visualization
- Marketing performance analysis

6. Data Cleaning and Preparation

Before performing the analysis, the dataset was examined to ensure data quality and consistency.

The following data cleaning activities were performed:

Duplicate Record Validation

The dataset was checked for duplicate campaign records.

Result:

No duplicate records were found.

Missing Value Validation

The dataset was checked for missing values across all columns.

Result:

No missing values were identified.

Numeric Data Validation

The following numerical columns were validated:

- Impressions
- Clicks
- Leads
- Conversions
- Revenue

- Acquisition Cost
- ROI

No invalid values were found.

Negative ROI values were retained because they represent campaigns where acquisition costs exceeded the revenue generated. These values provide meaningful business insights and therefore were not removed.

Date Validation

The Date column was verified to ensure it was stored in a valid date format for trend analysis and time-based visualizations.

Final Dataset

The cleaned dataset was successfully imported into Microsoft Power BI for dashboard development.

7. DAX Measures Created

To evaluate campaign performance dynamically, several DAX measures were created.

Measure	Purpose
Total Revenue	Total revenue generated
Total Impressions	Total audience reached
Total Clicks	Total campaign clicks
Total Leads	Total leads generated
Total Conversions	Total successful conversions
Average ROI	Average return on investment
Average Engagement Score	Average engagement level
CTR %	Click-through rate
Conversion Rate %	Percentage of clicks converted into successful outcomes

These measures automatically update based on user-selected filters.

8. Dashboard Development

An interactive Power BI dashboard was designed to provide a comprehensive overview of campaign performance.

The dashboard consists of:

KPI Cards

- Total Revenue
- Total Conversions
- Total Impressions
- Total Leads
- Total Clicks
- Average ROI
- Average Engagement Score
- Conversion Rate
- Click Through Rate (CTR)

Interactive Filters

Users can filter dashboard results by:

- Campaign Type
- Customer Segment
- Language
- Date

Visualizations

The dashboard includes:

- Revenue by Customer Segment
- Revenue by Campaign Type
- Top 10 Campaigns by Revenue
- Revenue Trend
- Engagement Score by Language
- Average ROI by Campaign Type

These visualizations allow marketing managers to evaluate campaign performance from multiple perspectives.

9. Dashboard Explanation

Each visualization answers an important business question.

Revenue by Customer Segment

Shows which customer groups generate the highest revenue and helps marketers prioritize high-value audiences.

Revenue by Campaign Type

Compares different campaign strategies to identify the most effective marketing approach.

Top 10 Campaigns by Revenue

Highlights the best-performing campaigns, enabling managers to understand which initiatives generated the greatest financial returns.

Revenue Trend

Displays changes in revenue over time, helping identify seasonal trends and campaign performance across different months.

Engagement Score by Language

Compares customer engagement across different campaign languages to evaluate audience response.

Average ROI by Campaign Type

Measures the profitability of different campaign strategies and helps determine which campaign types deliver the highest returns.

10. Key Insights

- The marketing campaigns generated approximately **\$28.7 billion** in total revenue, demonstrating the significant financial impact of marketing activities.
- Influencer and Social Media campaigns generated the highest overall revenue, indicating that digital and influencer marketing strategies contribute substantially to business growth.
- The **Working Women** customer segment generated the highest revenue among all customer groups, making it one of the most valuable target audiences.
- The dashboard reveals that the top-performing campaigns contribute a significant portion of the total revenue, suggesting that a relatively small number of campaigns drive a large share of business performance.
- Customer engagement scores remain relatively consistent across all campaign languages, indicating balanced audience interaction regardless of language preferences.
- The average ROI remains relatively similar across campaign types, suggesting that while revenue differs slightly, overall campaign profitability is fairly consistent across marketing strategies.

11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase investment in high-performing campaign types such as Influencer and Social Media campaigns, as they generate the highest revenue.
- Design personalized marketing campaigns targeting the **Working Women** customer segment, which demonstrates strong revenue potential.
- Analyze the characteristics of the top-performing campaigns and apply similar strategies to future marketing initiatives.
- Continue monitoring campaign ROI regularly to ensure marketing budgets are allocated efficiently and underperforming campaigns are optimized.

- Maintain multilingual marketing campaigns since engagement levels remain consistent across different languages, allowing businesses to effectively reach diverse customer groups.

12. Conclusion

This project successfully demonstrates how Microsoft Power BI can be used to transform marketing campaign data into meaningful business insights.

Through systematic data cleaning, DAX calculations, and interactive dashboard development, the project provides marketing managers with a comprehensive view of campaign performance, customer engagement, revenue generation, and profitability.

The dashboard enables data-driven decision-making by identifying high-performing campaign types, valuable customer segments, and revenue trends. It also highlights opportunities to optimize marketing strategies and improve future campaign performance.

Overall, this project showcases practical skills in data preparation, business intelligence, data visualization, marketing analytics, and dashboard development using Microsoft Excel and Microsoft Power BI.